MINISTRY OF EDUCATION AND VOCATIONAL TRAINING

CERTIFICATE OF SECONDARY EDUCATION EXAMINATION

URBAN WEST REGION – ZANZIBAR

FORM FOUR (IV) MOCK EXAMINATION – 2025

BASIC MATHEMATICS

(For Both School and Private Candidates)

041

TIME: 3: 00 HOURS **MONDAY 16TH JUNE 2025**

INSTRUCTIONS:

1. This paper consists of sections **A** and **B** with a total of **fourteen (14)** questions.
2. Answer **all questions** in section **A** and **B.** Each question in section **A** carries **six (6) marks** while each question in section **B** carries **ten (10) marks.**
3. **All** necessary working and answers for each question must be shown clearly.
4. **NECTA** mathematical tables and non-programmable Calculator **may be used.**
5. Write your **Examination Number** on every page of your answer booklet(s)
6. Cellular phone and unauthorized materials are **not allowed** in the examination room.

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SECTION A:

Answer all questions in this section

1. (a) A wall is 120 cm by 270 cm. I want to cover it with square tiles. What is the largest tile I can use so that I don't have to cut any?

(b) (i) In a mixed school of 880 pupils, $^3/_5$ are girls. Of the boys, $^4/_{11}$ study biology. How many boys do not study biology?

(ii) A farmer has 140 hectares on his farm. He uses 35 hectares to grow crops. What percentage of the farm is not used to grow crops?

2. (a) Solve these equations:

(i) $4^{8x-3} = 8^{5x+1}$

(ii) $\log_2(x+5) = \log_2 x + 2$

(b) Rationalize the denominator and simplify $\frac{\sqrt{2}}{\sqrt{3} - \sqrt{2}}$

3. (a) Out of 20 farmers, 15 had sheep and 12 had goats. 10 had both. How many:

(i) Had neither animal?

(ii) Had goat only?

(iii) Either animal?

(b) A jar contains 2 black marbles, 3 blue marbles and 1 white marble. If one marble is drawn and put aside and then another drawn and put aside, determine:

(i) The probability that both marbles drawn are black.

(ii) The number of black marbles to be added to the jar to make the probability of drawing 2 black marbles equal to $^1/_7$

4. (a) Given vectors $\underline{a} = 2i + 3j$, $\underline{b} = 3i + 7j$, and $\underline{c} = 11j$. Find scalars p and q such that $\underline{c} = p\underline{a} + q\underline{b}$.

(b) ΔABC has vertices A(1, 5), B(4, -1), and C(-2, -4). Show that it's a right-angled triangle and find its area.

5. (a) The areas of two similar triangles are in the ratio 2:3. If the altitude of the smaller triangle is 15cm, find the altitude of the larger triangle.

(b) (i) Find the circumference of a circle that inscribes a regular pentagon of perimeter 72cm.

(ii) A kite has an area of 12cm². If it's longer diagonal is 5cm more than the shorter diagonal. Find the length of the shorter diagonal.

6. (a) A man is paid 10,000 to build 5 animal huts at the same rate. How much will he be paid to build 16 huts?

(b) The resistance R ohm's of an electrical wire varies with the length L cm and inversely with the square of the radius r cm. If the radius is 0.1cm and the length is 20cm, the resistance is 0.4 ohms. Find R in terms of r and L.

7. Given:
Opening Stock 1.1.2023 is 34,430/=
Closing Stock 31.12.2023 is 26,720/=
Net purchases during 2023 is 219,290
Gross profit mark up 50%
Expenses for year is 45,880/=
Using the above transactions, find the following:

(i) Cost of goods sold

(ii) Sales

(iii) Net profit

(iv) Gross profit

8. (a) Write down the first 5 terms of the following sequence governed by the following rules:

(i) $An = 3^n + n$

(ii) $An = n(n+1)(n+2)$

(b) (i) If $n, n+1, n-1$ are in G.P, find the value of n. Hence, find the 4th term.

(ii) The fourth term of an A.P is twice as great as the 3rd term and the 12th term is 30. Find the first term and common difference.

9. (a) A rope of length 18cm is tied to the top of a flagpole. The other end of the rope is fixed to a point 13cm from the base of the flagpole. How high is the flagpole?

(b) In triangle ABC, if $\angle C = 90°$, AB = 5cm, BC = 4cm, and AC = 3cm. Find sin B and cos A, as well as angle A and angle B.

10. (a) (i) Solve for from x from $(3+x)(14-3x) = 30$.

(ii) Solve for x by using completing the square $3x^2 - 2x - 1 = 0$.

(b) A father is 28 years old than his son. If the sum of their ages is 54, find the son's age.

SECTION B:

Answer all questions in this section

11. (a) In the Final Examination of a Certain School, the Scores of Students in Commerce Subject were grouped as shown in the table:

Scores	65 - 69	70 - 74	75 - 79	80 - 84	85 - 89	90 - 94	95 - 99
Cumulative Frequency	10	22	43	49	58	62	66

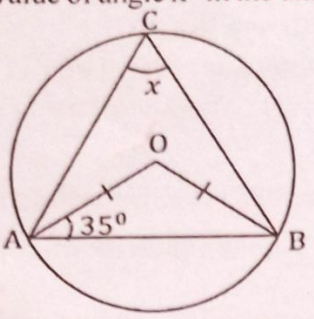
Using the information given in the table:

(i) Find the mean score correct to 2 decimal places, given an assumed mean of 82.

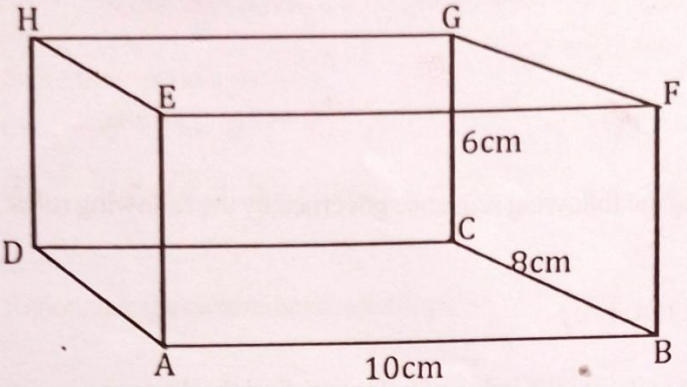
(ii) Calculate the median score.

(iii) Calculate the mode of the scores correct to 3 decimal places.

(b) Find the value of angle x° in the diagram given below that $\angle O\hat{A}B = 35°$.



12. (a) In the figure below shows a rectangular prism in which AB = 10cm, BC = 8cm and CG = 6cm.



(i) Calculate the length $\overline{BH}$ or $\overline{AG}$. Correct in 2 decimal places.

(ii) Calculate the angle that $\overline{AG}$ makes with the plane ABCD.

(b) (i) Find the shortest distance (in kilometers) between two points along a parallel of latitude 60°N, one is on longitude 30°E and other is 15°W. (Radius of the Earth is 6400km, Use $\pi = 3.14$)

(ii) A and B are any two points on the equator with longitude 40°W and 25°E respectively. Calculate the shortest distance of the two points in kilometers along the equator between A and B. (Use $R_E = 637\text{km}, \pi = 3.14$)

13. (a) (i) Find the value of x and y given that:

$$\begin{bmatrix} x & 4 \\ 3 & y \end{bmatrix} + \begin{bmatrix} 3x & 5 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} 8 & 9 \\ 0 & 9 \end{bmatrix}$$

(ii) The sum of two numbers x and is 140. If x is 34 greater than y. Write down two equations in x and y then solve them to find the two numbers by using the inverse matrix method.

(b) By using the matrix method find the image of the point (3,-2) after a reflection in the line $y = -x$ followed by another reflection in the line $x = 0$

(c) A translation takes point (5, 5) to the point (-7, -7) If it takes point (x,y) to (-4, -4) find the values of x and y.

14. (a) Show that the functions f and g are inverse of each other:

(i) $f(x) = 2 - 3x$ and $g(x) = \frac{2}{3} - \frac{x}{3}$

(ii) $f(x) = 4x - 5$ and $g(x) = \frac{1}{4}(x + 5)$

(b) A person requires 10, 12 and 12 units of chemical A, B and C respectively for his garden. A liquid product contains 5, 2 and 1 units of A, B and C respectively per jar and each jar is sold at Tsh 3000/ = A dry product contains 1, 2 and 4 units of A, B and C. respectively per carton and each carton is sold at the Tsh 2000/=. How many of each should he purchase to minimize the cost and still meet the requirement?